

CAMPINAS, Brazil

WMO: 837210

Lat: 23.007S

Lon: 47.135W

Elev: 661

StdP: 93.63

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 9.0	(c) 10.8	(d) 2.9	(e) 5.1	(f) 21.8	(g) 4.9	(h) 5.8	(i) 21.4	(j) 12.3	(k) 15.9	(l) 10.9	(m) 16.6	(n) 4.6	(o) 120	(p) 0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 9.8	(c) 33.8	(d) 20.2	(e) 32.8	(f) 20.5	(g) 31.8	(h) 20.6	(i) 23.8	(j) 28.6	(k) 23.3	(l) 28.2	(m) 22.9	(n) 27.8	(o) 3.8	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.1	24.8	22.1	18.2	24.4	21.8	17.9	24.2	75.0	29.1	73.0	28.2	71.4	27.9	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.0	(b) 10.1	(c) 9.2	(e) 6.3	(f) 35.8	(g) 2.2	(h) 1.3	(i) 4.7	(j) 36.7	(k) 3.4	(l) 37.5	(m) 2.2	(n) 38.2	(o) 0.6	(p) 39.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	24.5	24.9	24.2	22.7	19.6	18.7	18.7	20.1	21.5	22.9	23.2	24.3
	DBStd	3.22	2.02	1.86	1.74	2.14	2.39	2.40	2.62	2.74	2.94	2.73	2.25	1.85
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	92	0	0	0	1	14	23	28	16	7	2	1	0
	CDD10.0	4413	449	417	440	381	299	262	268	312	345	401	395	444
	CDD18.3	1462	191	183	182	132	55	35	38	69	102	144	146	186
	CDH23.3	10839	1316	1340	1241	891	332	235	315	652	915	1197	1063	1342
	CDH26.7	3708	488	519	428	258	48	18	41	180	363	499	373	494
(14) Wind	WSAvg	4.3	3.9	3.8	4.0	4.1	4.0	3.8	4.2	4.6	5.0	5.0	4.8	4.2
(15) Precipitation	PrecAvg	1235	231	185	144	47	53	36	34	26	63	99	127	170
	PrecMax	1538	353	350	323	119	113	182	133	59	178	174	199	271
	PrecMin	862	125	81	50	0	2	0	0	0	0	27	48	40
	PrecStd	153	71	61	67	30	28	46	33	20	50	43	48	58
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.1	33.0	31.9	29.2	28.2	29.1	31.9	34.2	35.2	33.2	33.8
		MCWB	21.5	20.4	22.1	20.7	19.4	17.7	17.0	17.4	18.5	19.7	20.5	20.6
	2%	DB	32.2	32.8	31.8	30.8	28.0	27.1	27.9	30.1	32.2	33.2	31.9	32.2
		MCWB	21.6	21.3	21.9	20.7	19.0	17.6	16.8	17.0	17.9	19.7	20.7	21.0
	5%	DB	31.1	31.2	30.2	29.2	26.8	26.0	26.8	28.9	30.8	31.8	30.2	31.0
		MCWB	21.6	21.5	21.6	20.5	18.4	17.4	16.6	16.8	17.8	19.7	20.4	21.1
	10%	DB	29.8	30.0	29.2	28.1	25.2	24.8	25.1	27.1	28.9	29.8	29.0	29.8
		MCWB	21.8	21.8	21.6	20.2	17.9	16.9	16.2	16.4	17.8	19.8	20.5	21.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	24.4	24.1	23.2	21.5	20.2	20.2	20.4	21.6	23.1	24.1	24.0
		MCDB	28.9	29.7	29.4	27.6	25.8	24.6	23.7	26.0	26.9	27.9	27.4	29.3
	2%	WB	23.6	23.6	23.3	22.5	20.4	19.2	18.7	19.1	20.5	22.1	23.0	23.2
		MCDB	28.3	28.6	28.4	27.2	25.0	23.6	23.6	24.8	26.2	28.3	27.5	28.2
	5%	WB	23.1	23.1	22.7	21.8	19.6	18.5	17.9	18.2	19.8	21.4	22.2	22.7
		MCDB	27.9	28.1	27.8	26.5	24.2	23.1	23.5	24.5	25.6	27.3	26.9	27.5
	10%	WB	22.6	22.6	22.2	21.3	18.9	17.7	17.1	17.4	19.1	20.7	21.6	22.1
		MCDB	27.1	27.3	27.1	25.6	23.1	22.5	22.4	23.8	24.6	26.1	26.1	26.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.8	9.8	10.2	10.1	10.7	11.4	12.4	11.7	11.2	10.5	10.1
		MCDBR	11.7	11.8	11.6	11.7	11.9	12.5	13.7	14.9	15.2	14.7	12.9	12.4
	5% WB	MCWBR	3.8	3.7	3.8	3.7	4.1	4.7	4.9	4.8	4.5	4.2	4.4	4.1
		MCWBR	9.5	10.0	10.1	9.9	10.2	10.3	11.2	12.3	11.6	11.3	10.1	9.8
(40) Clear-Sky Solar Irradiance	taub		0.403	0.404	0.386	0.365	0.346	0.351	0.364	0.399	0.486	0.448	0.428	0.411
		taud	2.420	2.397	2.452	2.482	2.509	2.483	2.427	2.301	2.048	2.224	2.293	2.369
	Ebn at Noon		942	927	917	894	871	844	843	846	810	876	913	936
		Edh at Noon	125	126	114	102	91	90	98	120	168	148	141	132
(44) All-Sky Solar Radiation	RadAvg		5.41	5.68	5.17	4.70	3.82	3.54	3.74	4.52	4.92	5.42	5.65	5.80
	RadStd		0.76	0.48	0.45	0.35	0.27	0.32	0.34	0.34	0.52	0.51	0.33	0.40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	-0.25	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	-0.30	-0.40	N/A	N/A	N/A

Nomenclature: See separate page